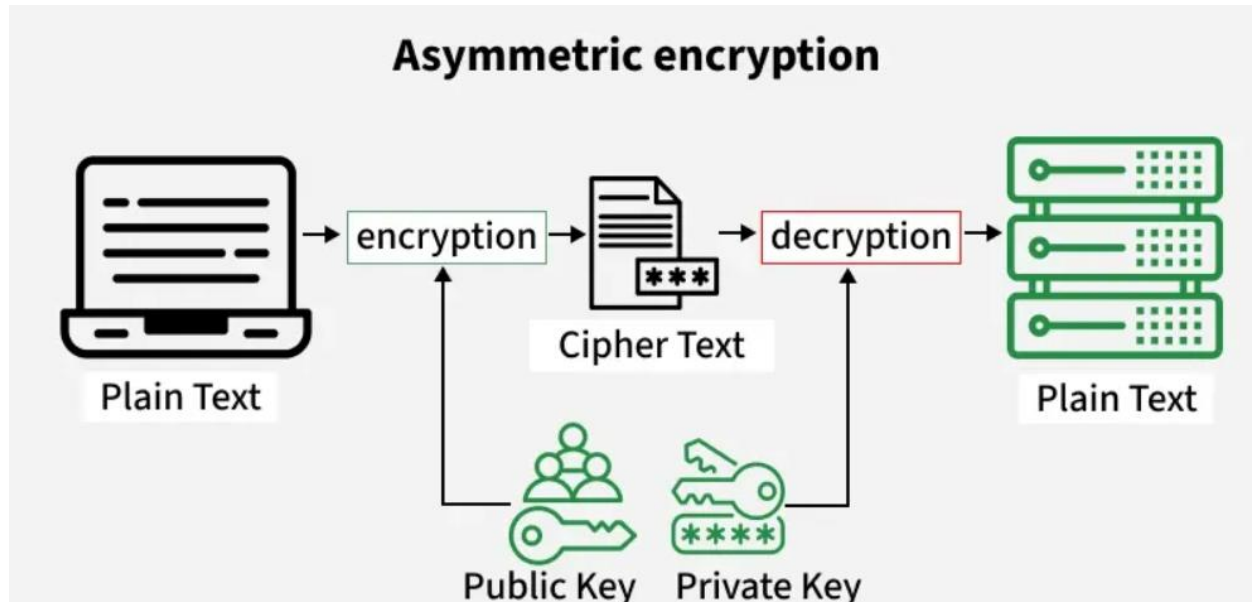


## Asymmetric Encryption

**Asymmetric Encryption** (also called **Public-Key Cryptography**) solved the single biggest flaw in security history: *how two people can communicate securely without meeting first to share a secret password.*

While symmetric encryption uses one single key to lock and unlock data, asymmetric encryption uses a mathematically linked **pair of keys**.



### The Key Pair: Public vs. Private

Whenever you set up asymmetric encryption, your computer generates two distinct cryptographic keys:

- **The Public Key:** This key is meant to be shared with the entire world. Anyone can use it to **encrypt** a message meant for you.
- **The Private Key:** This key is kept strictly secret on your device. Only this key can **decrypt** messages locked by your public key.

**The One-Way Lock Analogy:** Think of the public key as an open padlock. You hand open padlocks out to anyone who asks. If someone wants to send you a message, they put it in a box, snap your padlock shut, and send it to you. Once snapped shut, *not even the person who locked the box can open it again*. Only you hold the private key that fits the lock.

### Popular Asymmetric Algorithms

Asymmetric systems rely on complex mathematical problems that are easy to calculate in one direction but virtually impossible to reverse:

## 1. RSA (Rivest–Shamir–Adleman)

- **How it works:** Relies on the extreme difficulty of factoring the product of two massive prime numbers.
- **Key Sizes:** Requires large key sizes to stay secure (typically **2048-bit** or **4096-bit**).
- **Status:** The historic pioneer of public-key systems, still widely used, though gradually being replaced by faster alternatives.

## 2. ECC (Elliptic Curve Cryptography)

- **How it works:** Uses the complex algebraic properties of elliptic curves over finite fields.
- **Key Sizes:** Delivers the same military-grade security with much smaller keys (a **256-bit** ECC key offers roughly the same security as a **3072-bit** RSA key).
- **Status:** The modern standard for mobile devices, modern web browsers, and Bitcoin, because smaller keys mean faster calculations and lower power usage.

## Symmetric vs. Asymmetric: A Quick Comparison

| Feature          | Symmetric Encryption                                               | Asymmetric Encryption                                                  |
|------------------|--------------------------------------------------------------------|------------------------------------------------------------------------|
| Keys Used        | 1 Shared Secret Key                                                | 2 Keys (1 Public, 1 Private)                                           |
| Speed            | Extremely Fast                                                     | Slower (requires heavy math)                                           |
| Primary Use      | Bulk data transfer (encrypting drives, video streams, large files) | Key exchanges, digital signatures, and establishing secure connections |
| Key Distribution | Difficult (how do you share the key safely?)                       | Easy (public keys can be freely distributed)                           |

## The Two Killer Features

Asymmetric cryptography isn't just used for sending secret messages. It enables two fundamental pillars of the internet:

1. **Secure Key Exchange:** It allows two devices across the globe (like your browser and a bank server) to safely agree on a temporary symmetric key over an insecure network without hackers intercepting it.
2. **Digital Signatures:** If you flip the math and encrypt a document with your *private key*, anyone with your *public key* can verify it. This proves beyond doubt that the document came from you and hasn't been altered.